

Comprehensive Project Report: Type I Interferon Immunotherapy Response

Single-Cell RNA-Seq Analysis and Clinical Comparisons

Course/Project Title: Type I Interferon Immunotherapy Response: Single-Cell RNA-Seq Analysis

Prepared for: Student Project Guide (Project 3)

Author: Members of the 'Project 3' team **Date:** August 31, 2026

Abstract

This project replicates and critically evaluates the single-cell RNA-sequencing (scRNA-seq) pipeline modeling patient responses to anti-PD-1 immunotherapy based on baseline Type I Interferon (IFN-I) response capacity (IRC). Using a dataset of peripheral blood mononuclear cells (PBMCs) from healthy donors and cancer patients, we constructed a single-cell atlas comprising 94,513 cells. We calculated an individual-cell IRC module score using a six-gene interferon-stimulated protein (ISP) signature consisting of *BST2*, *EIF2AK2*, *ISG15*, *MX1*, *IFIT3*, and *IRF7*. Patient-derived cells were stratified into IRC-High and IRC-Low states using a cell-level median threshold, followed by differential expression and over-representation analysis (ORA) using GSEApY. Our computational pipeline successfully identified distinct transcriptional states corresponding to hyper- and hypo-interferon responsiveness. Comparing our findings with the primary literature (Boukhaled et al., *Nature Immunology*, 2022) revealed a critical clinical alignment: pre-therapy IFN-I priming of peripheral effector T cells (Teff) serves as an epigenetic poising mechanism that drives T-cell exhaustion and limits therapeutic response, while interferon resistance predefines longer overall survival.

1. Introduction and Primary Paper Reading Guide

1.1 The Clinical Paradox of Type I Interferons in Cancer Immunotherapy

Type I Interferons (IFN-I, including IFN- α and IFN- β) represent classic double-edged swords in tumor immunology. On one hand, acute, transient IFN-I signaling is vital for initiating innate and adaptive anti-tumor immune responses. It expands antigen-specific T cells, promotes dendritic cell (DC) cross-priming, and sustains cytotoxic T lymphocyte (CTL) effector functions.

On the other hand, chronic or sustained IFN-I signaling drives severe immunosuppressive feedback loops. It induces the expression of inhibitory ligands and factors, such as programmed death ligand-1 (PD-L1), programmed cell death protein 1 (PD-1), indoleamine 2,3-dioxygenase 1 (IDO1), and interleukin-10 (IL-10) on both antigen-presenting cells (APCs) and T cells. This chronic signaling drives persistent T-cell activation toward "T-cell exhaustion," a dysfunctional state that facilitates tumor immune escape.

Consequently, an immunological paradox arises where the host's capacity to respond to IFN-I can either restrict or promote anti-tumor efficacy, directly dictating clinical responses to immune checkpoint blockade (such as anti-PD-1).

1.2 Rationale for Baseline Peripheral Blood Profiling

To resolve how the host's pre-existing immune state impacts therapeutic success, Boukhaled et al. profiled baseline (pre-treatment) peripheral blood mononuclear cells (PBMCs) rather than on-treatment tumor biopsies. This design is supported by several critical clinical and technical considerations:

1. **Undetectable and Unreliable Tissue Signaling:** During chronic diseases, circulating IFN- γ levels are generally undetectable, and tissue-level expression of interferon-stimulated genes (ISGs) varies widely across tissues and cells, making bulk tumor profiling highly variable and clinically elusive.

2. **Systemic Immunity Requirement:** Effective anti-PD-1 blockade relies heavily on the rejuvenation of systemic, peripheral blood immune responses rather than localized tumor-infiltrating lymphocytes alone.

3. **Clinical Stability of Basal Poising:** The authors demonstrated that the pre-therapy IFN- γ score (IS) was highly stable when comparing two baseline samples acquired a week apart, proving that peripheral blood provides a robust, non-invasive, and reproducible indicator of a patient's baseline physiological immune state.

4. **Disconnect from Intratumoral Inflammation:** Importantly, the pre-treatment peripheral effector T-cell (Teff) IRC did not correlate with baseline intratumoral inflammation (as measured by an 18-gene tumor-inflamed signature, TuIS), ruling out the possibility that low peripheral responsiveness was simply due to the physical compartmentalization of inflammatory T cells within the tumor microenvironment (TME). Instead, low peripheral IRC reflected an intrinsic capacity to respond dynamically and sustain interferon signaling *on-treatment*, preventing early refractory exhaustion.

2. Methods

Our single-cell transcriptomic workflow was implemented in Python using the **Scanpy** and **Anndata** libraries. Preprocessing, normalization, feature selection, dimensionality reduction, clustering, and pathway enrichment were executed sequentially.

2.1 Data Loading and Quality Control (QC)

Individual 10x Genomics PBMC matrices from 2 Healthy Donors (HD1, HD2) and 8 Cancer Patients (P1–P8) were loaded using Scanpy's `sc.read_10x_mtx()` function. Donor classifications were annotated by checking sample prefixes (samples starting with "HD" were classified as Healthy_Donor, and all others as Patient). Individual samples were concatenated into a single merged dataset containing **101,020 cells and 36,601 genes**.

We defined three quality-control gene categories using prefix string matching: - **Mitochondrial Genes:** Genes prefixed with MT-. - **Ribosomal Genes:** Genes prefixed with RPS or RPL. - **Hemoglobin Genes:** Genes matching HBA, HBB, HBD, HBE, HBG, HBM, HBQ, or HBZ.

Using `sc.pp.calculate_qc_metrics()`, cellular QC metrics were calculated. Low-quality, dying, or contaminated cells were excluded using the following rigorous filtering thresholds: - Number of detected genes (`n_genes_by_counts`) ≥ 300 . - Mitochondrial transcript percentage (`pct_counts_mt`) $< 15\%$. - Hemoglobin transcript percentage (`pct_counts_hb`) $< 5\%$.

Genes detected in fewer than **5 cells** were removed to exclude extremely sparse or uninformative features.

After the QC filtering the final output was a merged dataset of **94,513 cells and 31,054 genes**.

2.2 Normalization and Highly Variable Gene (HVG) Selection

To account for differences in library sizes, raw transcript counts were normalized to a target sum of 10,000 counts per cell using `sc.pp.normalize_total()`. To stabilize variance, normalized counts were log-transformed using $x' = \log(1+x)$ via `sc.pp.log1p()`. The complete normalized and log-transformed expression matrix was preserved in `adata.raw` to allow visualization of canonical lineage markers and interferon-stimulated genes that might be excluded during highly variable feature selection.

Highly variable genes (HVGs) were identified using `sc.pp.highly_variable_genes()`, selecting the top **2,500 genes**. We incorporated the `batch_key="sample"` parameter to ensure that feature selection captured biological variation while controlling for technical sample-specific differences. Due to the massive scale of the dataset (94,513 cells), the dataset was explicitly subsetted to these 2,500 HVGs *before* executing computationally intensive downstream regression.

Technical and unwanted sources of variation, specifically `total_counts` (sequencing depth) and `pct_counts_mt` (mitochondrial content), were regressed out using `sc.pp.regress_out()`.

Finally, gene-wise scaling was performed using `sc.pp.scale()` to standardize expression to zero mean and unit variance, with values clipped at a maximum absolute value of 10 to minimize the influence of extreme outliers during principal component analysis.

2.3 Dimensionality Reduction, Clustering, and Lineage Annotation

Principal Component Analysis (PCA) was performed on the scaled 2,500 HVG matrix, computing the first 50 principal components. An elbow plot of the variance ratio was evaluated, and a cell-cell neighborhood graph was constructed using the first **30 principal components** and **20 nearest neighbors** per cell (`sc.pp.neighbors()`). A 2D visualization was generated using Uniform Manifold Approximation and Projection (UMAP).

Leiden community detection was applied at a resolution of **1.0** to identify transcriptionally distinct cellular clusters. Major circulating immune lineages were annotated by evaluating canonical marker expression across the Leiden clusters using the full normalized matrix in `adata.raw`: - **CD4+ T Cells**: *CD3D*, *CD4*, *IL7R*, *CCR7* - **CD8+ T Cells**: *CD3D*, *CD8A*, *CD8B*, *GZMB* - **Natural Killer (NK) Cells**: *NKG7*, *GNLY*, *NCAM1* - **B Cells**: *MS4A1*, *CD19*, *CD79A* - **Monocytes**: *CD14*, *FCGR3A*, *LYZ* - **Dendritic Cells (DCs)**: *FCER1A*, *CST3*

Visualizations included Leiden-colored UMAPs, marker-specific feature plots, and a systematic marker expression dotplot. Because several canonical marker genes were not included among the selected highly variable genes, marker expression was also examined using the full normalized expression matrix retained in `adata.raw`. Marker expression was visualized using dot plots across Leiden clusters and feature plots on the UMAP embedding.

2.4 Quantification of Interferon Response Capacity (IRC)

The baseline Type I Interferon Response Capacity (IRC) was quantified using the six core interferon-stimulated protein (ISP) genes established in the primary literature: ***BST2***, ***EIF2AK2*** (***PKR***), ***ISG15***, ***MX1***, ***IFIT3***, and ***IRF7***. Because these biological markers are not guaranteed to reside within the mathematically selected 2,500 HVGs, the module score was computed using

the full normalized expression matrix in `adata.raw` via Scanpy's `sc.tl.score_genes()` function with the `use_raw=True` parameter.

Computational Note: Rather than calculating a simple arithmetic mean of the six genes (which is highly sensitive to background noise and library depth), Scanpy's `score_genes` algorithm calculates a relative module score. It identifies the average expression of the target six-gene signature and subtracts the average expression of control gene sets. These control sets consist of genes randomly selected from bins of similar overall expression levels across the entire dataset, thereby successfully isolating specific interferon-stimulated transcriptional changes from basal cellular metabolic activity.

2.5 Cell-Level Stratification and Differential Expression

To investigate the biological differences between interferon-responsive states, the concatenated AnnData object was subsetting to contain only cancer patient cells (`donor_type == "Patient"`). The cell-level median IRC score among all patient cells was calculated. Patient cells were stratified into two transcriptionally distinct populations: - **IRC-High Patient Cells:** Cells with an IRC score $\geq$ the patient median. - **IRC-Low Patient Cells:** Cells with an IRC score $<$ the patient median.

Differential expression (DE) between the IRC-High and IRC-Low cell populations was computed using the Wilcoxon rank-sum test implemented in `sc.tl.rank_genes_groups()`. Genes were ranked, and p-values were adjusted using the Benjamini-Huang multiple testing correction. The complete DE table was saved as `IRC_High_vs_Low_DE.csv`.

2.6 Over-Representation Pathway Enrichment Analysis (ORA)

To evaluate the biological processes enriched in the IRC-High state, we isolated significantly upregulated genes from the differential expression results using the following filters: - *log2foldchange* > 0.5 (equivalent to a 1.41-fold expression increase). - *adjusted p-value* < 0.05 .

The top **100 genes** matching these filters were selected as the input gene list for over-representation analysis (ORA) using GSEAPy's `gp.enrichr()` interface. Enrichment was tested against three major curated databases: 1. **Reactome 2022** 2. **KEGG 2021 Human** 3. **GO Biological Process 2023**

Enriched pathways were ranked using Benjamini-Hochberg adjusted p-values, and the complete enrichment table was exported to `pathway_enrichment_results.csv`.

3. Results (Figures and Analysis)

Figure 1: Quality Control and Filtering Assessment

Quality control metrics were rigorously evaluated to ensure that cellular datasets were free of low-quality, apoptotic, or red blood cell contaminants.

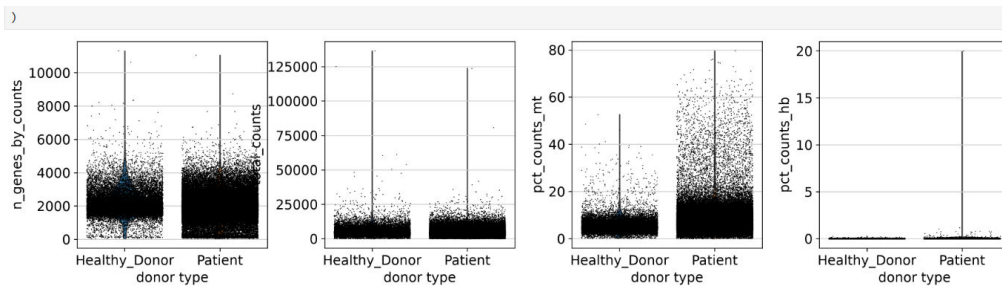


Figure 1A: Raw QC Metrics by Donor Type (Pre-Filtering)

Violin plots displaying the raw cellular distribution of detected genes (`n_genes_by_counts`), total transcript counts (`total_counts`), mitochondrial transcript percentage (`pct_counts_mt`), and hemoglobin transcript percentage (`pct_counts_hb`) compared between healthy-donor (HD) and cancer patient cohorts

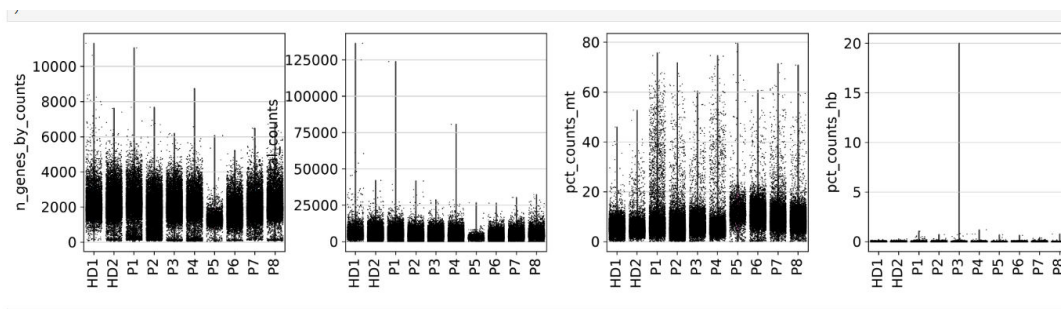


Figure 1B: Raw QC Metrics by Sample (Pre-Filtering)

Violin plots showing the raw distribution of the same four QC metrics across individual samples to monitor batch-to-batch technical consistency.

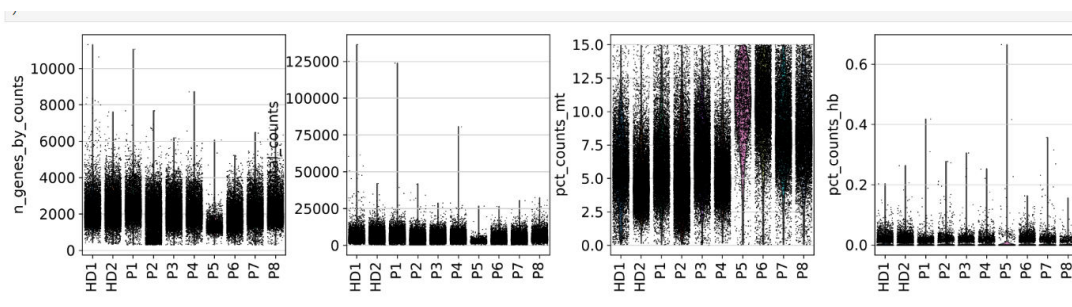


Figure 1C: QC-Filtered Metrics by Sample (Post-Filtering)

Violin plots showing the distribution of the same metrics across individual samples after applying the filtering thresholds of `n_genes_by_counts` ≥ 300 , `pct_counts_mt` $< 15\%$, and `pct_counts_hb` $< 5\%$, demonstrating successful elimination of low-quality and contaminated cells.

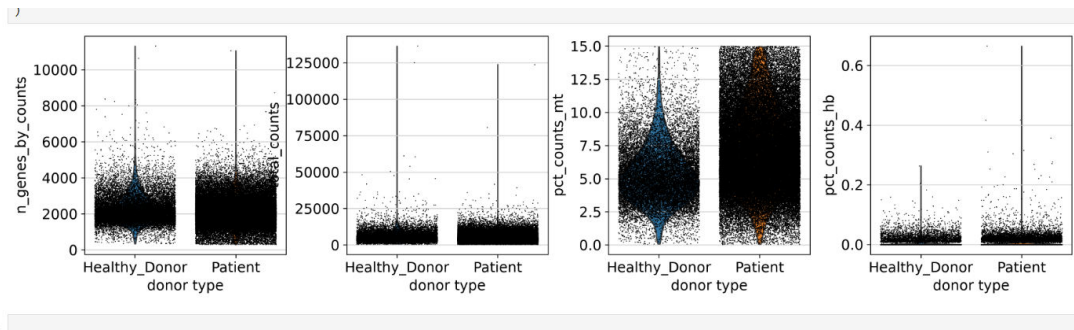


Figure 1D: QC-Filtered Metrics by Donor Type (Post-Filtering)

Violin plots showing the filtered QC distributions compared between donor groups.

Preprocessing & Highly Variable Gene (HVG) Selection Plot

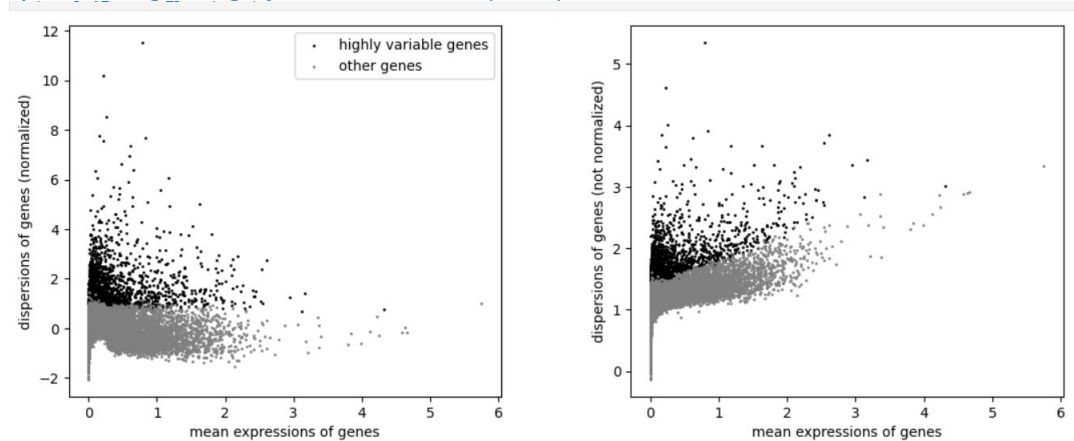


Figure 1E: Preprocessing & Highly Variable Gene (HVG) Selection Plot

Dispersion vs. Mean expression scatter plots demonstrating highly variable gene (HVG) selection, where the top 2,500 genes with the most informative biological variation are highlighted in black.

Metric	Pre-Filtering Dataset	Post-Filtering Dataset	Summary of Removed Components
Cell Count	101,020 cells	94,513 cells (or student's exact count)	Apoptotic cells (high mitochondrial genes $\geq 15\%$) and lysed cell debris (<300 detected genes)
Gene Count	36,601 genes	31,054 genes	Sparse genes detected in fewer than 5 individual cells

Table 1A: Table showcasing the pre-filtering and post-filtering states of cell and gene counts

	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_ribo	pct_counts_hb
count	101020.000000	101020.000000	101020.000000	101020.000000	101020.000000
mean	2024.614344	4746.331543	7.841968	3.282089	0.002825
std	773.474282	2904.851562	5.931536	3.074897	0.064613
min	37.000000	41.000000	0.000000	0.000000	0.000000
25%	1609.000000	3111.000000	4.638738	1.301058	0.000000
50%	1967.000000	4207.000000	6.572817	2.324680	0.000000
75%	2385.000000	5746.000000	9.338955	4.199184	0.000000
max	11307.000000	136340.000000	79.525482	53.267567	20.000000

Table 1B: Table displaying the descriptive statistics of the pre-filtered data

	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_hb
sample				
HD1	2033.0	4412.0	5.821004	0.0
HD2	2068.0	4194.0	4.636642	0.0
P1	2245.0	5049.0	6.135209	0.0
P2	1909.0	4022.0	5.011136	0.0
P3	1996.5	4317.0	6.621499	0.0
P4	1984.0	4513.0	5.272609	0.0
P5	1298.0	2545.0	11.416914	0.0
P6	1678.0	3454.5	10.859177	0.0
P7	1960.5	3965.5	8.808832	0.0
P8	1970.0	4327.0	8.525476	0.0

Table 1C: Table displaying the qc-summary of the pre-filtered data

[7]

	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_ribo	pct_counts_hb
count	94513.000000	94513.000000	94513.000000	94513.000000	94513.000000
mean	2102.129633	4943.688477	6.772288	3.092707	0.002480
std	712.498918	2858.925781	3.018094	2.719732	0.010673
min	301.000000	334.000000	0.032321	0.000000	0.000000
25%	1670.000000	3273.000000	4.539125	1.259061	0.000000
50%	2003.000000	4320.000000	6.328233	2.232262	0.000000
75%	2420.000000	5880.000000	8.715163	4.012794	0.000000
max	11306.000000	136339.000000	14.998391	53.267567	0.664820

Table 1D: Table displaying the descriptive statistics of the post-filtered data

	n_genes_by_counts	total_counts	pct_counts_mt	pct_counts_hb
sample				
HD1	2041.0	4434.0	5.791857	0.0
HD2	2084.0	4253.0	4.600699	0.0
P1	2308.5	5291.0	5.791340	0.0
P2	1939.0	4133.0	4.923367	0.0
P3	2016.0	4389.5	6.554592	0.0
P4	2009.5	4619.0	5.147215	0.0
P5	1344.0	2633.0	10.542797	0.0
P6	1711.5	3528.5	10.365805	0.0
P7	1990.0	4053.0	8.474369	0.0
P8	1990.0	4393.0	8.346796	0.0

Table 1E: Table displaying the qc-summary of the post-filtered data

Figure 2: Dimensionality Reduction and Unsupervised Clustering

Unsupervised Leiden community detection at a resolution of 1.0 was performed on the cell-cell neighborhood graph, revealing the structured, high-dimensional landscape of the immune cells.

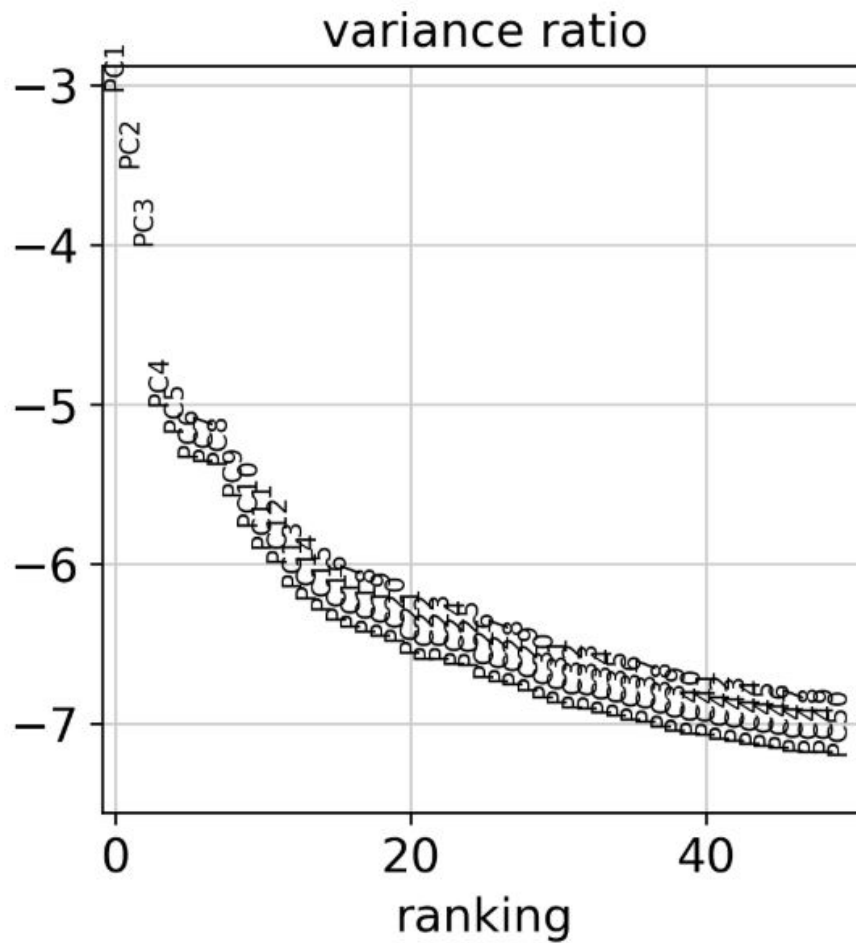


Figure 2A: PCA Variance Ratio (Elbow Plot)

PCA elbow plot illustrating the percentage of variance explained by each of the first 50 principal components, validating the selection of the first 30 PCs for cell-cell neighborhood graph construction

Figure 2B: UMAP Embeddings of Resolved Atlas (Leiden, Donor Type, Sample)

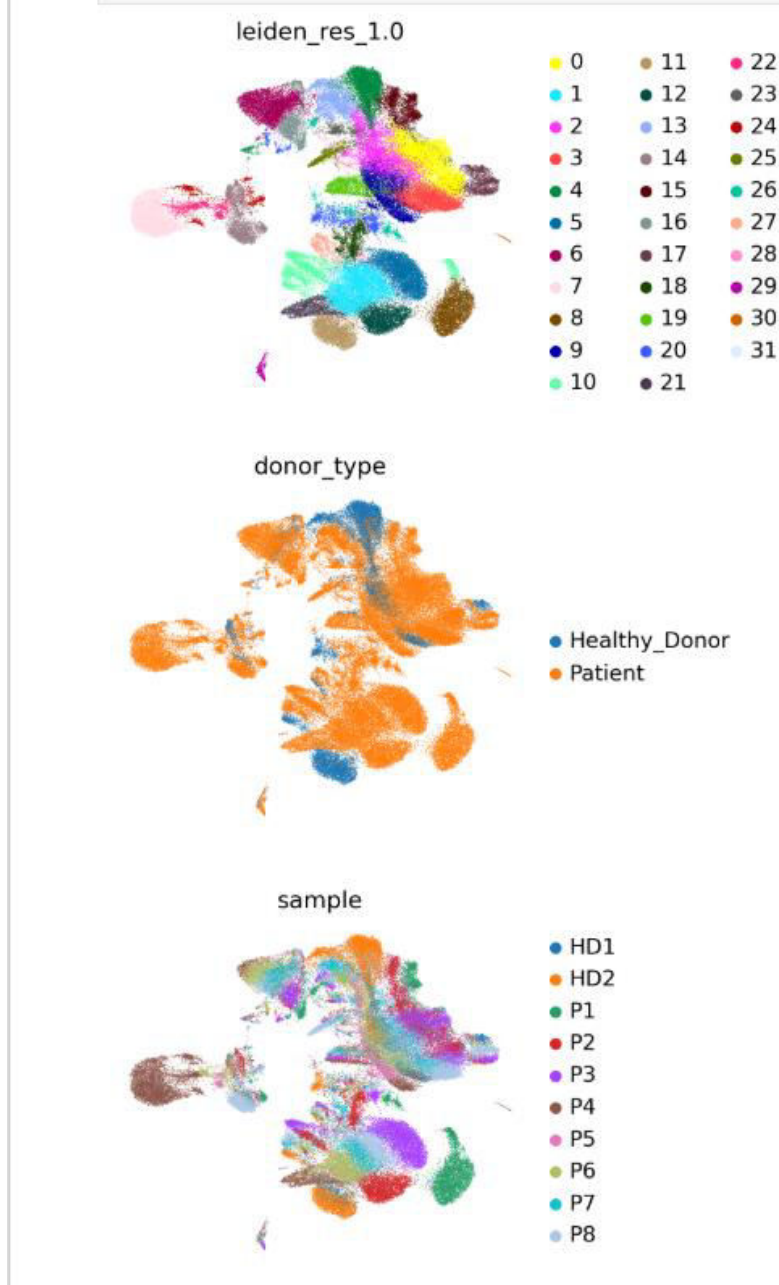


Figure 2B: UMAP Embeddings of Resolved Atlas (Leiden, Donor Type, Sample)

Two-dimensional UMAP embedding of 94,513 PBMCs colored by their unsupervised Leiden cluster assignments at a resolution of 1.0 (resolving transcriptionally distinct clusters 0–31) (top), colored by donor type (middle), and colored by individual sample (bottom), verifying well-integrated samples with minimal batch effects.

Figure 2C: UMAP Feature Plots of Canonical Lineage Markers in HVG subset

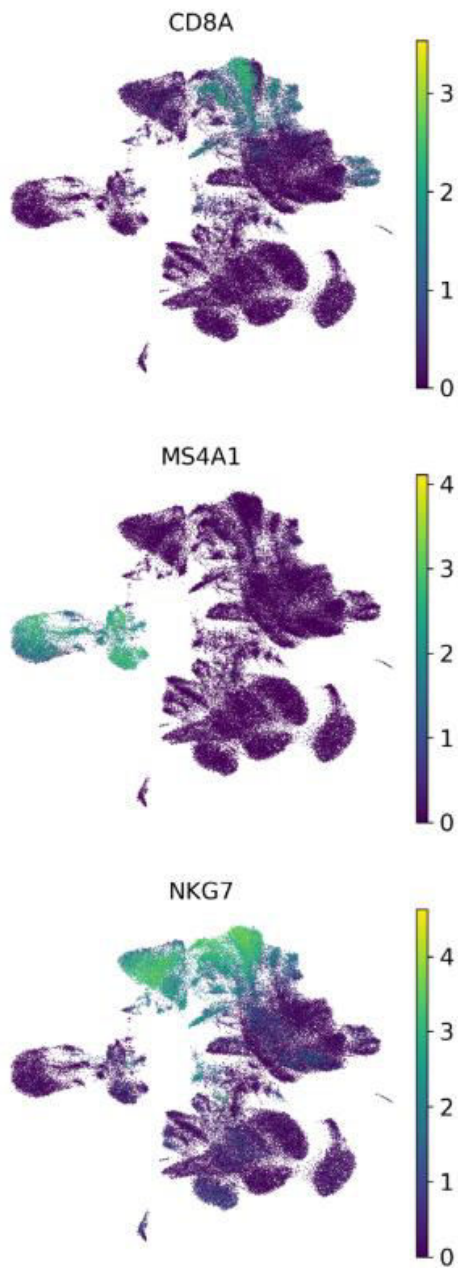


Figure 2C: UMAP Feature Plots of Canonical Lineage Markers in HVG subset

Feature plots showing the single-cell expression of canonical lineage-defining markers CD8A, MS4A1, and NKG7 across the UMAP embedding to identify CD8+ cytotoxic T cells, B cells, and NK cells, respectively.

Figure 2D: UMAP Feature Plots of Canonical Lineage Markers in combined normalized data

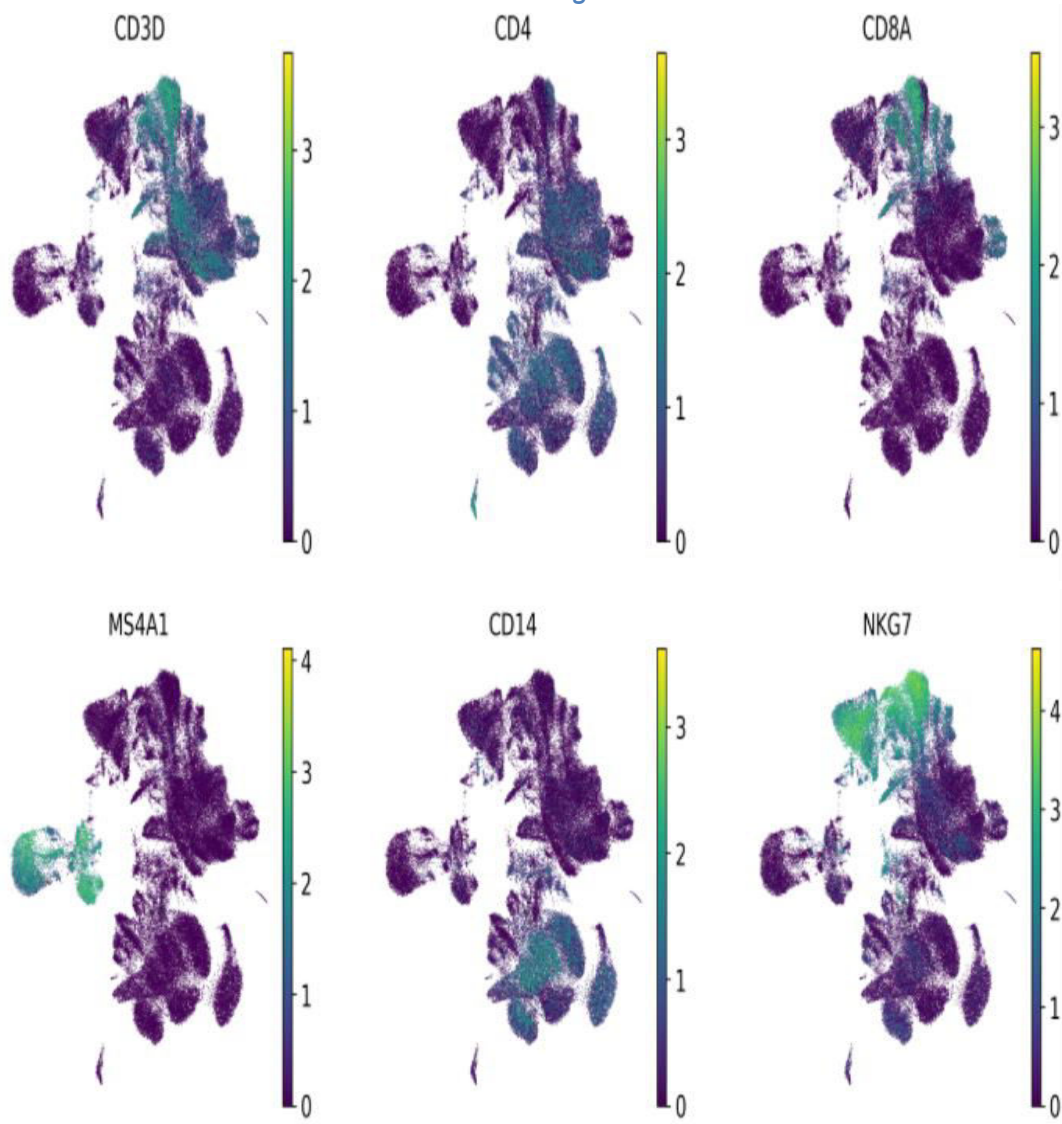


Fig 2D: UMAP Feature Plots of Canonical Lineage Markers in combined normalized data

Corresponding feature plots showing the single-cell expression of canonical lineage-defining markers. Canonical lineage markers like *CD3D*, *CD4*, *IL7R*, and *CCR7* are absent from the highly variable gene subset.

Figure 3: Canonical Lineage Characterization

To validate the biological identity of our Leiden clusters, canonical immune-lineage markers were evaluated simultaneously across all clusters.

Figure 3A: Comprehensive Marker Dotplot Utilizing Full Expression Matrix (adata.raw)

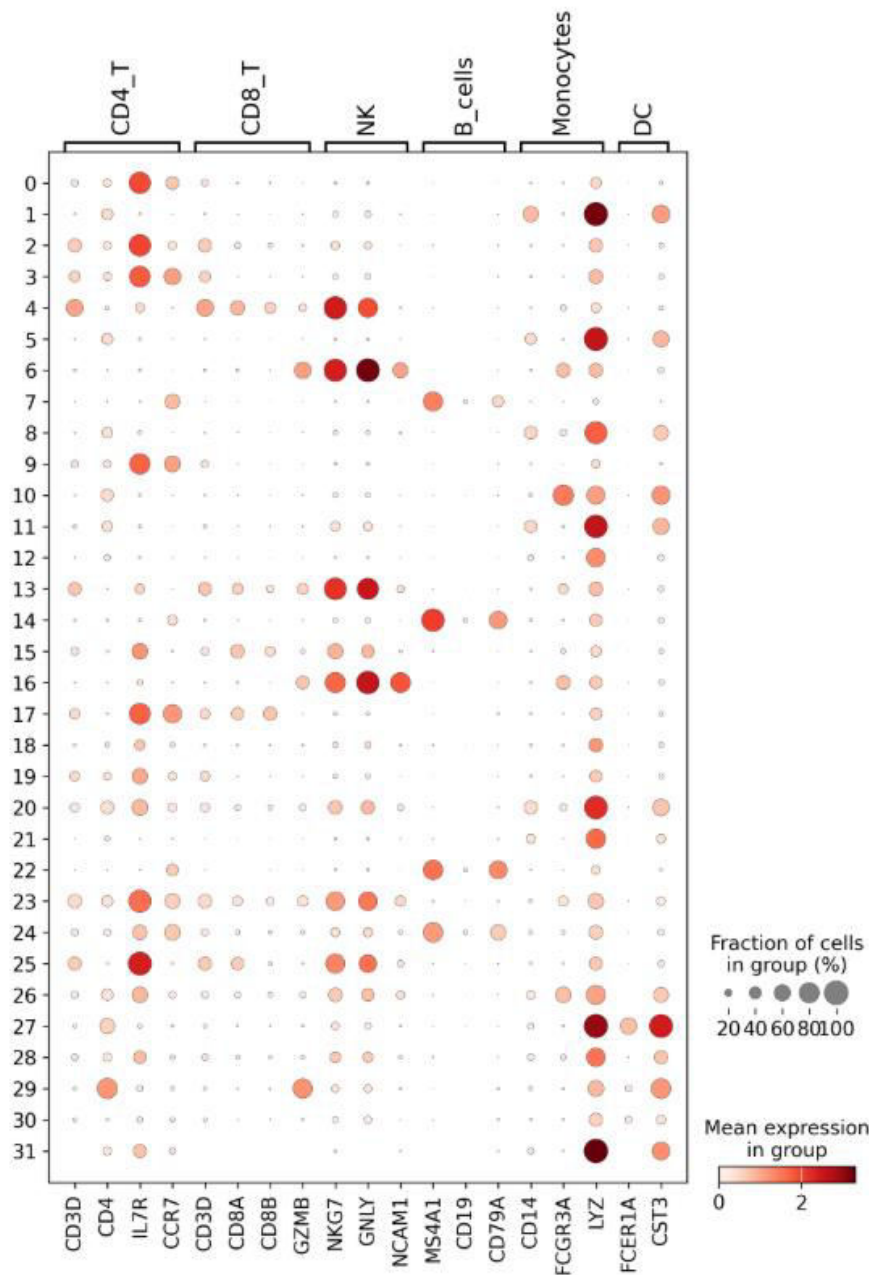


Figure 3A: Comprehensive Dotplot of Canonical Lineage Markers

Systematic dotplot mapping the expression of canonical lineage-defining markers across Leiden clusters 0–31 using the full normalized matrix in adata.raw. Dot size reflects the percentage of cells expressing the marker within each cluster, and color intensity represents log-normalized, scaled mean expression. Canonical markers evaluated include T-cell markers (*CD3D*, *CD4*,

IL7R, *CCR7*, *CD3E*, *CD8A*, *CD8B*, *GZMB*), NK-cell markers (*NKG7*, *GNLY*, *NCAM1*), B-cell markers (*MS4A1*, *CD19*, *CD79A*), Monocyte markers (*CD14*, *FCGR3A*, *LYZ*), and Dendritic Cell markers (*FCER1A*, *CST3*)

Figure 3B: Restricted Marker Dotplot Utilizing Highly Variable Genes (*adata.X*)

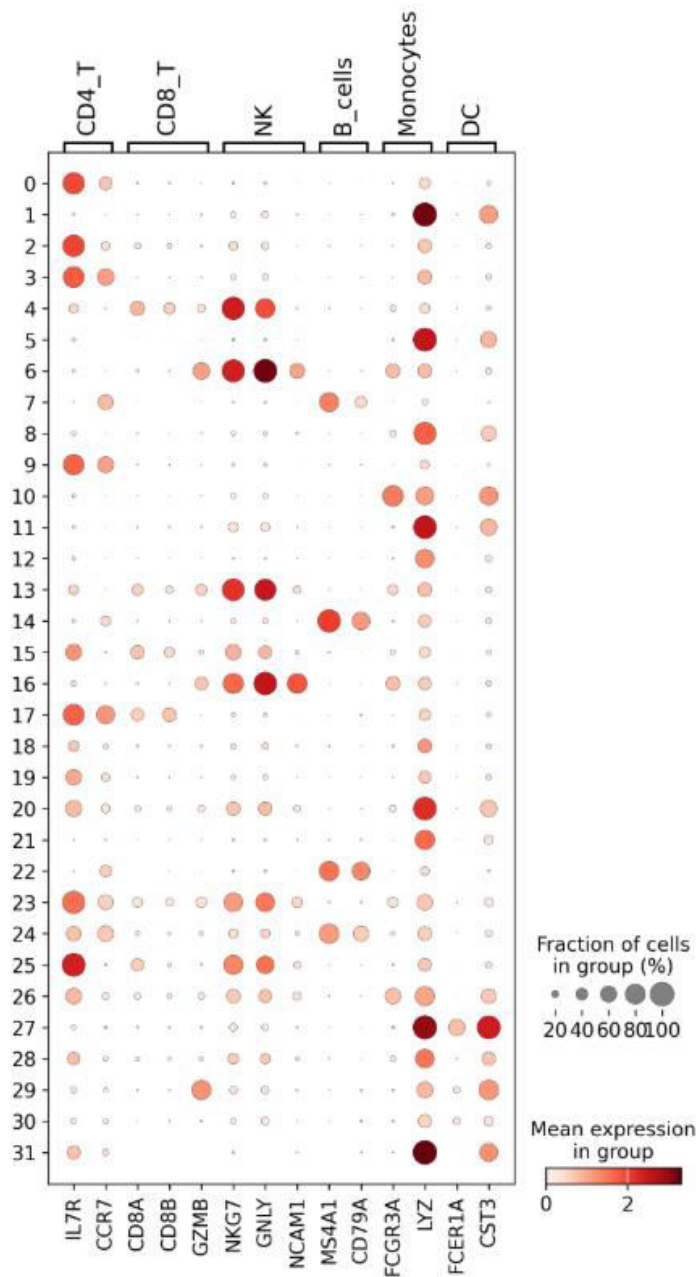


Figure 3B: Restricted Dotplot demonstrating missing markers in HVG subset

Corresponding dotplot using only genes in *adata.X* (restricted to the 2,500 HVGs). This comparison demonstrates a critical single-cell preprocessing pitfall: canonical lineage markers like *CD3D*, *CD4*, *IL7R*, and *CCR7* are absent from the highly variable gene subset because their baseline variation does not meet the mathematical variance threshold, highlighting the

necessity of preserving raw log-transformed matrices in adata.raw for biologically accurate cell annotation.

Figure 4: Interferon Response Capacity (IRC) Module Score Distribution

The six-gene interferon-stimulated signature was quantified per-cell, projecting host responsiveness onto identified lineages.

Figure 4A: Baseline IRC score UMAP projection and cohort comparison

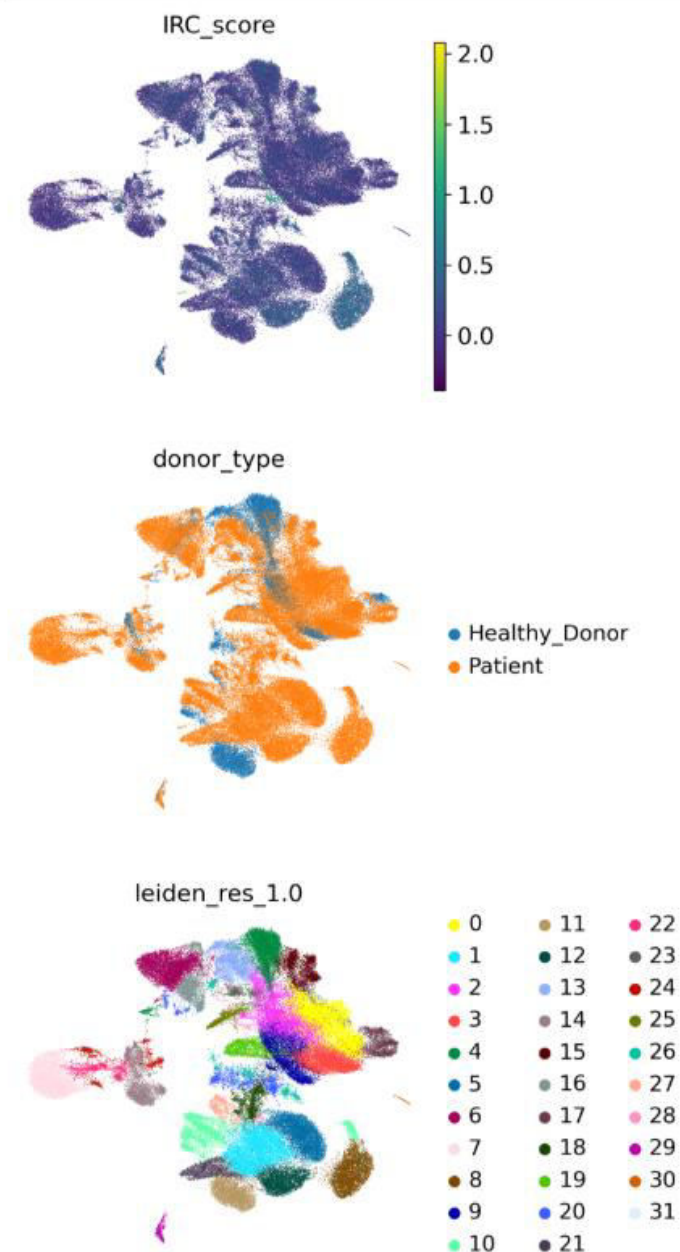


Figure 4A: Baseline IRC score UMAP projection and cohort comparison

Aligned UMAP visualizations colored by the relative continuous IRC module score (top), donor type (middle), and Leiden cluster (bottom), illustrating localized regions of baseline interferon

priming. The score is calculated using the six core interferon-stimulated genes (*BST2*, *EIF2AK2*, *ISG15*, *MX1*, *IFIT3*, *IRF7*) on the adata.raw matrix. Myeloid compartments (Monocytes and DCs) exhibit the highest basal IRC scoring, while lymphoid lineages remain quiescent

Figure 4B: Continuous IRC Score Distribution Across Individual Samples

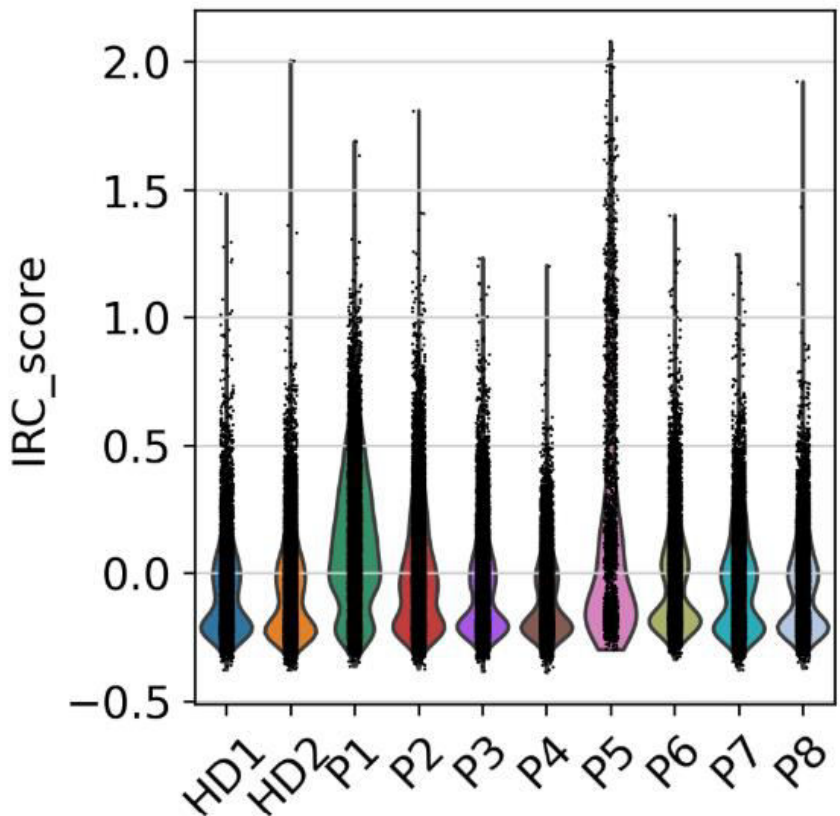


Figure 4B: Continuous IRC Score Distribution Across Individual Samples

Violin plot depicting the cellular distribution of IRC scores across individual healthy-donor and patient samples (HD1, HD2, P1–P8), showing substantial sample-to-sample heterogeneity in baseline interferon responsiveness.

Figure 5: Differential Gene Expression and Pathway Remodeling

Figure 5A: Differential gene expression between IRC-High and IRC-Low patient-derived cells.

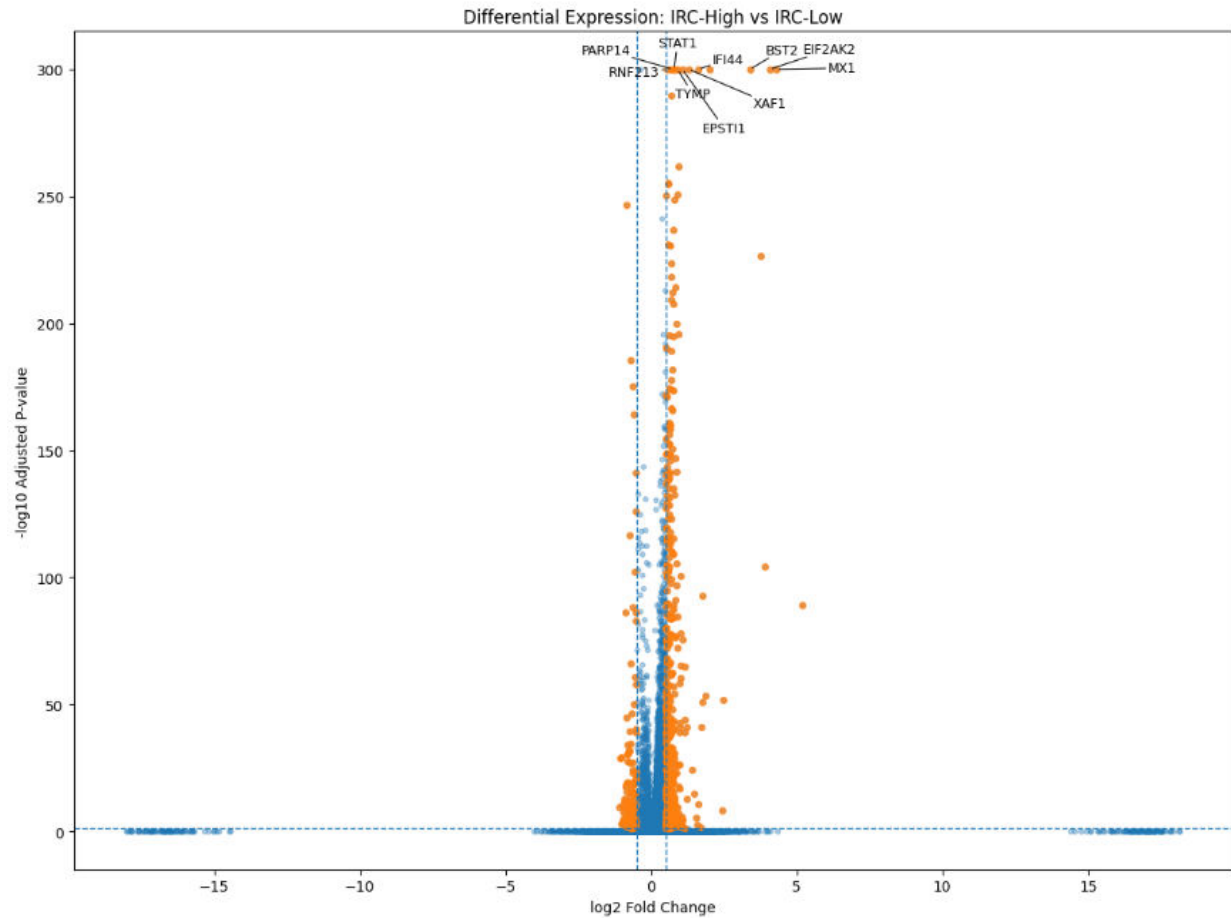


Figure 5A: Differential gene expression between IRC-High and IRC-Low patient-derived cells.

Volcano plot showing the differential expression of genes between cells classified as IRC-High and IRC-Low based on their IRC scores. The x-axis represents $\log_2$ fold change, while the y-axis represents $-\log_{10}(\text{adjusted } P\text{-value})$. Genes exceeding the predefined thresholds of $|\log_2 FC| > 0.5$ and adjusted $P\text{-value} < 0.05$ were considered differentially expressed. The ten most statistically significant genes are labelled.

The resulting volcano plot demonstrated distinct transcriptional differences between the two IRC states, with several genes showing increased expression in IRC-High cells. The ten most statistically significant genes are highlighted in Figure 5A.

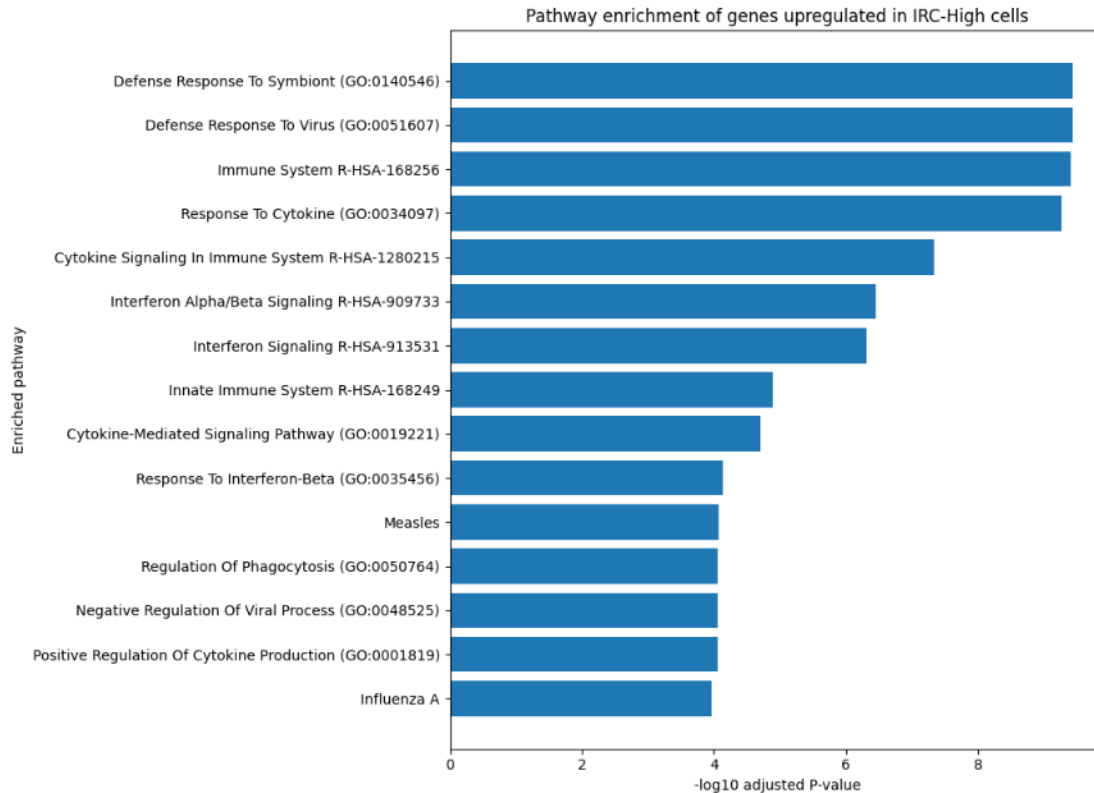


Figure 5B: Pathway enrichment of genes upregulated in IRC-High cells.

Bar plot showing the most significantly enriched biological pathways among genes upregulated in the IRC-High group. Pathway enrichment was performed using Enrichr with Reactome, KEGG and Gene Ontology Biological Process gene sets. The x-axis represents $-\log_{10}(\text{adjusted P-value})$, with larger values indicating stronger statistical enrichment. Notably, interferon alpha/beta signaling, interferon signaling, cytokine signaling, innate immune processes, and antiviral responses were among the significantly enriched pathways

4. Discussion (Analytical and Critical Review)

This study used a single-cell RNA-sequencing dataset of peripheral blood mononuclear cells (PBMCs) to investigate cellular heterogeneity and basal type-I interferon (IFN-I) response capacity in the context of anti-PD-1 immunotherapy. The analysis followed a Scanpy-based workflow involving quality-control filtering, normalization, identification of highly variable genes, regression of technical variation, scaling, principal component analysis (PCA), neighborhood graph construction, UMAP visualization, Leiden clustering, immune-lineage marker analysis, and calculation of an IFN-I Response Capacity (IRC) score. Together, these analyses provide a framework for examining how basal interferon-associated transcriptional states vary across circulating immune cells.

PBMC atlas and immune-cell heterogeneity

The first objective was to determine whether the PBMC dataset could be separated into biologically meaningful immune populations. After quality-control filtering and dimensionality reduction, the cells were organized into distinct Leiden clusters based on their transcriptional similarity. The UMAP representation provided a two-dimensional visualization of these relationships, while canonical immune markers were used to evaluate the biological identity of the clusters.

The marker analysis included established lineage-associated genes such as **CD3D/CD4** for T-cell populations, **CD8A/CD8B** for CD8-associated T cells, **NKG7/GNLY** for NK cells, **MS4A1/CD19/CD79A** for B cells, **CD14/FCGR3A/LYZ** for monocytes, and **FCER1A/CST3** for dendritic-cell-associated populations. These markers provided the basis for interpreting the transcriptional differences observed between Leiden clusters. Thus, the clustering analysis demonstrates the cellular heterogeneity expected in a PBMC atlas and provides a framework for examining IFN-I activity in different immune compartments.

Quantification of IFN-I response capacity

The second objective was to quantify basal IFN-I responsiveness at single-cell resolution. An IRC score was calculated using the six-gene interferon-stimulated protein (ISP) signature identified in the reference study: **BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7**. Importantly, the score was calculated using the normalized and log-transformed expression values stored in `adata.raw`. Therefore, the IRC calculation was **not restricted to the 2,500 highly variable genes** used for PCA and clustering; the six signature genes were assessed from the full normalized expression matrix retained in `adata.raw`.

The resulting IRC score provides a relative measure of the interferon-associated transcriptional state of each cell. Visualization of IRC scores on the existing UMAP showed how this state was distributed throughout the cellular landscape, while violin plots demonstrated differences in IRC-score distributions between samples. This approach therefore converts the qualitative observation of interferon-stimulated gene expression into a quantitative single-cell measurement.

Lineage-specific IFN-I responsiveness

The third research question concerned whether IFN-I responsiveness differs between immune-cell populations. The IRC score was visualized together with the PBMC cellular structure to examine the distribution of interferon-associated activity across the dataset. The biological interpretation of these differences should be made in conjunction with the lineage-marker results rather than from IRC score alone, because a high score indicates an interferon-associated transcriptional state but does not independently identify the underlying cell type.

The observed IRC-associated transcriptional program is consistent with the biological role of IFN-I in activating interferon-stimulated genes and antiviral and inflammatory responses. However, the present analysis should not be interpreted as demonstrating that one immune lineage is intrinsically more responsive to IFN-I in a functional sense. The IRC score measures transcriptional activity of the selected signature genes and therefore represents a molecular proxy for IFN-I response capacity.

IRC-high versus IRC-low states

The fourth research question examined transcriptional differences between cells with relatively high and low IRC scores. Patient-derived cells were separated at the median IRC score into

IRC-High and IRC-Low groups. Differential-expression analysis was then performed using the Wilcoxon rank-sum test.

The resulting analysis identified genes associated with the IRC-High state, providing evidence that cells with greater IRC scores exhibit broader transcriptional differences beyond the six genes used to calculate the score. This is important because it suggests that the IRC signature reflects a wider interferon-associated cellular state rather than simply reproducing the expression of the six input genes.

An important limitation is that the present implementation classified **individual patient cells**, rather than assigning an IRC-High or IRC-Low status to entire patients. Consequently, the differential-expression results should be described as a comparison of IRC-High versus IRC-Low **patient-derived cells**, rather than as a direct comparison of IRC-High versus IRC-Low patients. In addition, the analysis was not restricted specifically to effector T cells. Therefore, although the results provide evidence for IRC-associated transcriptional differences, they cannot by themselves establish that high IRC in effector T cells predicts clinical response to anti-PD-1 therapy.

Pathway remodeling associated with IRC

The fifth research question addressed the molecular pathways associated with the IRC-High state. Significant genes upregulated in IRC-High cells were selected using a threshold of log2 fold change > 0.5 and adjusted $p < 0.05$, with the top 100 genes subsequently subjected to pathway enrichment analysis using Reactome, KEGG, and Gene Ontology Biological Process databases.

The enriched pathways were strongly related to immune and interferon biology. Among the most significant results were **Immune System**, **Cytokine Signaling in Immune System**, **Interferon Alpha/Beta Signaling**, **Interferon Signaling**, **Innate Immune System**, and **ISG15 Antiviral Mechanism**. Gene Ontology analysis additionally identified processes such as **defense response to virus**, **response to cytokine**, **cytokine-mediated signaling**, and **response to interferon-beta**. These results provide independent pathway-level support for the interpretation that the IRC-High state represents a broader interferon-associated immune program.

The enrichment results are particularly relevant to the biological context of the study because IFN-I signaling has a context-dependent role in anti-tumor immunity. IFN-I can promote immune surveillance through induction of interferon-stimulated genes, enhancement of antigen presentation, and activation of immune-response programs.

At the same time, persistent interferon signaling can be accompanied by regulatory and adaptive-resistance mechanisms. The present analysis demonstrates the presence of an interferon-associated transcriptional state, but it does not directly establish these downstream immunosuppressive effects. Therefore, the paradoxical nature of IFN-I should be considered as biological context rather than as a direct conclusion from the present dataset.

Direct comparison with the reference study

The present findings are broadly consistent with the central biological premise of the reference study, which proposed that baseline IFN-I response capacity in immune cells may be associated

with subsequent responsiveness to PD-1 blockade. In the present analysis, the six-gene IRC signature consisting of **BST2, EIF2AK2, ISG15, MX1, IFIT3, and IRF7** was used to quantify the interferon-associated state of individual cells.

Cells with higher IRC scores showed enrichment of broader immune and interferon-related transcriptional programs, including interferon-alpha/beta signaling, interferon signaling, cytokine signaling, innate immune processes, and antiviral responses. This supports the interpretation that the IRC signature captures a biologically meaningful IFN-I-associated state rather than simply reflecting expression of isolated genes.

The reference study places particular emphasis on **IFN-I priming in effector T cells** and its potential relationship with anti-PD-1 responsiveness. The biological rationale is that an appropriately primed immune compartment may be better positioned to mount an effective response following release of PD-1-mediated inhibition. Our findings provide molecular evidence for the presence of such interferon-associated states at single-cell resolution; however, the current analysis does not establish the clinical predictive relationship demonstrated or proposed in the reference study.

There are several important differences between the present analysis and the reference study. First, our IRC-High versus IRC-Low differential-expression analysis was performed at the **cell level among patient-derived cells**, rather than by assigning individual patients to IRC-High or IRC-Low groups. Second, the differential-expression analysis was not restricted specifically to effector T cells. Finally, the present workflow did not directly test treatment response, progression, survival, or another clinical outcome following anti-PD-1 therapy. Therefore, our results should be interpreted as supporting the **biological component** of the study's hypothesis rather than independently validating IRC as a predictive biomarker of anti-PD-1 response.

Nevertheless, the strong enrichment of interferon and immune-response pathways in IRC-High cells is compatible with the reference study's proposed model of baseline immune priming. The results suggest that cells exhibiting a higher basal IFN-I transcriptional state are characterized by coordinated activation of interferon-stimulated, cytokine, antiviral, and innate immune programs. Whether this state translates into improved therapeutic responsiveness depends on cellular context and cannot be determined from the present analysis alone.

Overall, the present study therefore reproduces an important molecular aspect of the reference study: **IRC captures a measurable basal IFN-I-associated immune state at single-cell resolution**. The findings provide supporting evidence for the biological plausibility of using IFN-I response capacity as a potential biomarker in immunotherapy research, while also highlighting the need for patient-level, effector-T-cell-specific, and clinical-outcome analyses to determine whether IRC can reliably predict anti-PD-1 responsiveness.

6. References

1. Boukhaled, G. M. et al. Pre-encoded responsiveness to type I interferon in the peripheral immune system defines outcome of PD1 blockade therapy. *Nature Immunology*, 23(8), 1273–1283 (2022). [67, 70]
2. Ayers, M. et al. IFN- γ -related mRNA profile predicts clinical response to PD-1 blockade. *J. Clin. Invest.*, 127, 2930–2940 (2017). [110, 123]
3. Cristescu, R. et al. Pan-tumor genomic biomarkers for PD-1 checkpoint blockade–based immunotherapy. *Science*, 362, eaar3593 (2018). [111, 123]
4. Bengsch, B. et al. Epigenomic-guided mass cytometry profiling reveals disease-specific features of exhausted CD8 T cells. *Immunity*, 48, 1029–1045 (2018). [117, 125]
5. Scanpy: Large-scale single-cell gene expression analysis. *Genome Biology*, 19, 15 (2018). [1]
6. AnnData: Storage and manipulation of single-cell datasets. *Bioinformatics*, 37(13), 1930–1931 (2021). [1]

7. Appendix: AI Usage Disclosure

7.1 Tool and Model Disclosure

- **AI Tool:** ChatGPT, Gemini Notebook (assisted by the underlying reasoning and development agent).
- **Core Role:** Assisted in report structuring, methods documentation, biological results translation, and critical methodology comparison between student and paper workflows. Commenting on codes and generate codes for volcano plots and bar plot for pathway enrichment as there was no code in the guideline for this.

7.2 Prompt and Purpose Log

Purpose of Submitted Prompt	Functionality
Give me code blocks of the script with comments	To understand the codes clearly
ok tell me what to run to generate the barplot and volcano plot	To generate the figure 5
Generate me the submittable paper structure with my findings	The methods section and result section structure in the paper
Discuss the findings based on the given rules	The discussion section

7.3 Student Verification Statement

All AI-assisted outputs, descriptions, and structural elements were independently reviewed, cross-referenced with the source documents (01_data_loading_qc.docx through 05_pathway_enrichment.docx, what I did.docx, project3_guide.pdf, and paper.pdf), and validated for factual accuracy. Parameter choices and statistical interpretations have been fully validated, and we assume complete academic authorship over this report.